

Preparation of the Zinc Enolate Equivalent of Amides by Zinciomethylation of Isocyanates: Catalytic Asymmetric Reformatsky-Type Reaction

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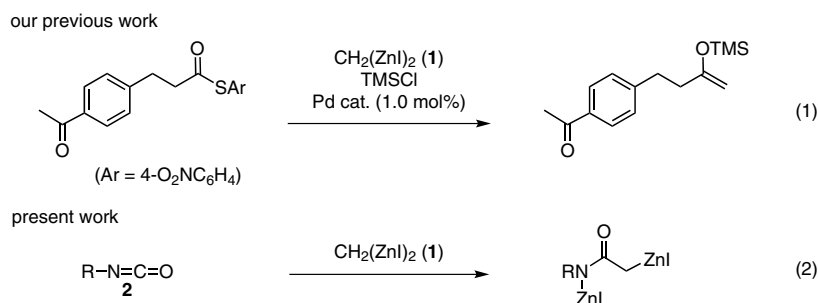
Abstract: Bis(iodozincio)methane [$\text{CH}_2(\text{ZnI})_2$] transforms isocyanates ($\text{R}-\text{N}=\text{C}=\text{O}$) into the enolate equivalent of amides via zinciomethylation. The reactivity of the enolate equivalent as a nucleophile toward aldehydes depends on the R group of the isocyanate. For the enolate equivalent formed from phenyl isocyanate, the addition of a catalytic amount of an optically active amino alcohol, which acts as an activator, leads to a catalytic asymmetric Reformatsky-type reaction.

Key words: organozinc reagents, dizinc compounds, Reformatsky reaction, asymmetric synthesis, catalysis

Organozinc reagents have been widely used as versatile nucleophiles, which possess reasonable reactivity and high compatibility with various functional groups.¹ Some classical methods, such as metal–zinc exchange, deprotonation, and halogen–zinc exchange, have been utilized to prepare these reagents. These methods are problematic, as the conditions used are often too harsh for some functional groups in the target organozinc species. However, the use of complex reagent systems has changed the situation dramatically. For example, it has been shown that the insertion of zinc into organic halides is facilitated in the presence of a stoichiometric amount of lithium chloride.² Amide zincates³ and the zinc–magnesium–lithium complex base, $(\text{tmp})_2\text{Zn} \cdot 2 \text{MgCl}_2 \cdot 2 \text{LiCl}$,⁴ have been shown to be useful reagents to provide heteroarylzinc compounds by deprotonation. These reagents have led to a renewed interest in organozinc species as an indispensable tool for organic synthesis.⁵

To increase the value of organozinc reagents as synthetic tools, a completely different strategy for the preparation of these reagents may be significant. We have studied the reactivity of bis(iodozincio)methane (**1**),⁶ which is easily obtained from diiodomethane and zinc powder.⁷ A number of specific molecular transformations, based on its reactivity as a *gem*-dimetal species, have been shown.⁸ Among these transformations, a zinciomethylation reaction can be focused on as a new method for the preparation of organozinc species.⁹ For example, a palladium-catalyzed cross-coupling reaction of iodobenzene with **1** gave a benzylzinc reagent. Moreover, treatment of acylating reagents with **1** gives α -iodozinciocarbonyl compounds, which can tautomerize into the corresponding enolates. In fact, treatment of a thioester with dizinc **1** and trimethylsilyl chloride in the presence of a palladium catalyst gave the corresponding silyl enolate via Fukuyama coupling [Scheme 1, (1)].¹⁰ In this study, we focus on the use of isocyanates as electrophiles [Scheme 1, (2)]. We discuss the synthesis and reactivity of the zinc enolate equivalent of amides, which can be prepared by the addition of dizinc **1** to isocyanates **2**.¹¹

Various isocyanates were treated with dizinc **1** (Table 1). A tetrahydrofuran solution of **1** (0.13 M, 1.5 mL) was added to a solution of the isocyanate **2** (0.2 mmol) in solvent (4 mL). The resulting reaction mixture was quenched with a saturated aqueous solution of ammonium chloride. At 25 °C, benzoyl isocyanate (**2a**, R = Bz) gave the corresponding amide **3a** and the diamide **4a**; the diamide **4a** is



Scheme 1 Homologative zinc enolate preparation: (1) cross-coupling reaction of a thioester with **1**; (2) addition of **1** to isocyanates **2**

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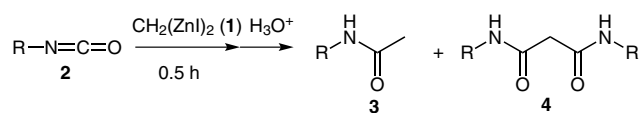
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the adduct of the initial zinciomethylated product with another equivalent of isocyanate **2a** (Table 1, entry 1). Lowering the reaction temperature ($-60\text{ }^{\circ}\text{C}$), to prevent the formation of **4a** gave **3a** selectively (Table 1, entry 3). Aryl and alkyl isocyanates **2b–e** were less electrophilic than **2a** in the reaction with dizinc **1**. To complete the addition of **1** to phenyl isocyanate (**2b**), a higher reaction temperature ($80\text{ }^{\circ}\text{C}$) was required (Table 1, entries 4 and 5). The use of toluene as a less polar solvent did not affect the yield of **3b** (Table 1, entry 6). The addition of substituents to the aryl isocyanates to tune the electron density on the benzene ring resulted in a slight decrease in the yield (Table 1, entries 7 and 8). An alkyl (cyclohexyl) isocyanate also gave the corresponding amide **3e** at $80\text{ }^{\circ}\text{C}$ (Table 1, entry 9).

The addition of benzaldehyde in place of the aqueous workup was examined to perform a Reformatsky-type reaction. The corresponding enolate equivalent formed from benzoyl isocyanate (**2a**) showed high nucleophilicity toward benzaldehyde (**5a**), even at $-40\text{ }^{\circ}\text{C}$, and gave the adduct **6aa** in 90% yield (Table 2, entry 1). In the case of the enolate equivalent formed from phenyl isocyanate (**2b**), a higher reaction temperature ($140\text{ }^{\circ}\text{C}$) was required to give the corresponding adduct **6ba** in reasonable yield (Table 2, entry 5). The other isocyanates, **2c–e**, also gave the corresponding adducts **6ca–ea** in poor yields at $25\text{ }^{\circ}\text{C}$ (Table 2, entries 6–8).

The enolate equivalent adduct of dizinc **1** and an isocyanate **2** has three possible forms, **7a–c** (Scheme 2). Although these structures are in equilibrium, the contribution of each structure depends on the R group. ^1H

Table 1 Reaction of Dizinc **1** with Isocyanates **2**^a



Entry	R	Solvent	Temp (°C)	Yield ^b (%) of 3	Yield ^b (%) of 4
1	Bz (2a)	THF	25	6	88
2	Bz (2a)	THF	0	32	62
3	Bz (2a)	THF	-60	73	16
4	Ph (2b)	THF	25	48	<1
5	Ph (2b)	THF	80	99	<1
6	Ph (2b)	toluene	80	99	<1
7	4-MeOC ₆ H ₄ (2c)	THF	80	88	<1
8	4-F ₃ CC ₆ H ₄ (2d)	THF	80	96	<1
9	Cy (2e)	THF	80	95	<1

^a Reaction conditions: dizinc **1** (0.13 M in THF, 1.5 mL, 0.2 mmol), isocyanate **2** (0.2 mmol), solvent (4 mL).

^b Isolated yields.

NMR analysis of the mixture resulting from the reaction of dizinc **1** and benzoyl isocyanate (**2a**) or phenyl isocyanate (**2b**) in tetrahydrofuran-*d*₈ gave some insight on the preferred structure of the formed enolate equivalents. The singlet peak at -1.3 ppm from the reaction of **1** and **2a** (R = Bz) suggests the formation of imine-like **7a**, while

Biographical Sketches



Ryosuke Haraguchi was born in 1988 and raised in Chiba, Japan. He obtained his B.Eng. degree from Kyoto University in 2009. He then started his master's

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Seiji Matsubara was born in 1959 and raised in Kobe, Japan. He obtained his B.Eng. degree in 1981 and M.Eng. degree in 1983 in the group of Prof. Hitosi Nozaki at Kyoto University. He obtained a Ph.D. degree from Kyoto University in 1986 under the direction of Prof. Hitosi Nozaki and Prof. Kiiro Utimoto. In 1984–1985, he also studied fluorine chemistry in the

group of Prof. Manfred Schlosser, Université de Lausanne, Switzerland. He joined Kyoto University as an assistant professor in Prof. Utimoto's group in 1986. After postdoctoral research with Prof. Barry M. Trost in 1988–1989 at Stanford University, USA, he became an associate professor at Kyoto University in 1995. In 2006, he became a full professor at Kyoto Uni-

versity. His research focuses on the preparation of cyclic compounds using organozinc reagents, transition-metal-catalyzed reactions, and organocatalytic reactions. He received the Inoue Research Award for Young Scientists in 1986 and the Incentive Award in Synthetic Organic Chemistry, Japan in 1998.

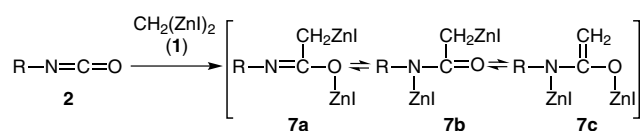
the singlet peak at 1.8 ppm from the reaction of **1** and **2b** (R = Ph) implies the formation of C-enolate **7b**.¹² The up-field chemical shift of the enolate equivalent from **2a** implies this compound has a stronger anionic character than that of the enolate equivalent from **2b**. The higher nucleophilicity observed for the **2a** enolate equivalent in the reaction with aldehydes can be understood in this way. As shown in Scheme 3, it is also plausible that enolate equivalent **7a**, formed from **2a**, may form chelated structures **8a** and **8b**. There is some enhancement of negative charge on the α -carbon atom in **8b**.

Table 2 Reformatsky-Type Reaction Starting from Dizinc **1** and Isocyanates **2**^a

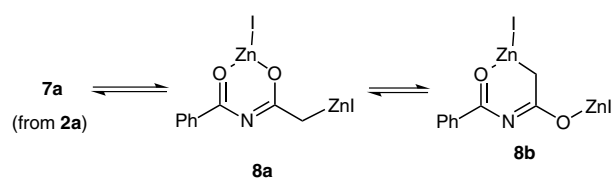
$\text{R}-\text{N}=\text{C}=\text{O} \xrightarrow[\text{T}_1, 0.5 \text{ h}]{\text{CH}_2(\text{ZnI})_2 \text{ (1)}} \text{PhCHO (5a)} \xrightarrow[\text{T}_2, 3 \text{ h}]{\text{H}_3\text{O}^+} \text{R}-\text{N}-\text{C}(=\text{O})-\text{CH}_2-\text{CH}(\text{OH})-\text{Ph}$					
Entry	R	Solvent	T ₁ (°C)	T ₂ (°C)	Yield ^b (%) of 6
1	Bz (2a)	THF	−60	−40	90 (6aa)
2	Ph (2b)	THF	80	−40	<5 (6ba)
3	Ph (2b)	THF	80	25	13 (6ba)
4	Ph (2b)	toluene	80	25	20 (6ba)
5	Ph (2b)	toluene	80	140	85 (6ba)
6	4-MeOC ₆ H ₄ (2c)	toluene	80	25	22 (6ca)
7	4-F ₃ CC ₆ H ₄ (2d)	toluene	80	25	39 (6da)
8	Cy (2e)	toluene	80	25	6 (6ea)

^a Reaction conditions: dizinc **1** (0.13 M in THF, 0.2 mmol), isocyanate **2** (0.2 mmol), solvent (4 mL), aldehyde **5a** (0.2 mmol).

^b Isolated yields.



Scheme 2 Possible structures of the adduct formed from dizinc **1** and an isocyanate **2**



Scheme 3 Possible chelation of the enolate equivalent **7a** formed from **1** and **2a**

Treatment of 1,1-diiodoethane with zinc powder in the presence of a lead catalyst also gives the corresponding *gem*-dizinc reagent, 1,1-bis(iodozincio)ethane (**9**).¹³ The

reaction of **9** and isocyanates **2** affords the zinc enolate equivalent of propionamide. As shown in Table 3, the formation of the enolate equivalent was evaluated by the yield of the protonated amide products.

Table 3 Reaction of Dizinc **9** with Isocyanates **2**^a

$\text{R}-\text{N}=\text{C}=\text{O} \xrightarrow[\text{0.5 h}]{\text{MeCH(ZnI)}_2 \text{ (9)}} \xrightarrow{\text{H}_3\text{O}^+} \text{R}-\text{N}-\text{C}(=\text{O})-\text{CH}_2-\text{CH}_3 \text{ (10)} + \text{R}-\text{N}-\text{C}(=\text{O})-\text{CH}(\text{CH}_3)-\text{C}(=\text{O})-\text{NH-R} \text{ (11)}$					
Entry	R	Solvent	Temp (°C)	Yield ^b (%) of 10	Yield ^b (%) of 11
1	Bz (2a)	THF	−20	53	<5
2	Bz (2a)	THF	−40	48	<5
3	Ph (2b)	THF	−60	53	<5
4	Ph (2b)	THF	25	<5	<5
5	Ph (2b)	THF	80	13	<5
6	Ph (2b)	toluene	80	22	<5

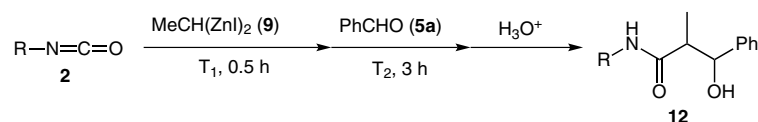
^a Reaction conditions: dizinc **9** (0.13 M in THF, 0.2 mmol), isocyanate **2** (0.2 mmol), solvent (4 mL).

^b Isolated yields.

The Reformatsky-type reaction was also examined for the enolate equivalents formed from dizinc **9** and isocyanates **2**. As shown in Table 4 (entry 1), the enolate equivalent formed from **9** and **2a** reacted with benzaldehyde (**5a**) to give the adduct **12aa** in 47% yield with a *syn/anti* diastereoselectivity of 6:1.¹⁴ The reactivity of the enolate equivalent formed from **9** and **2b** was too low to undergo the Reformatsky-type reaction with **5a** (Table 4, entries 2–4).

The low reactivity to aldehyde at 25 °C of the enolate equivalents of amides prepared from dizinc **1** and phenyl isocyanate (**2b**) or cyclohexyl isocyanate (**2e**) (Table 2, entries 4 and 8), implied the possibility of asymmetric induction of the Reformatsky-type reaction in the presence of a catalytic amount of an optically active activator. Although a variety of catalytic asymmetric alkylations of aldehydes using organozinc reagents have been developed,¹⁵ examples of the catalytic asymmetric Reformatsky-type reaction using a zinc enolate are limited.¹⁶ Several examples of the Reformatsky reaction using stoichiometric amounts of a chiral source¹⁷ and a few examples of catalytic asymmetric zinc enolate aldol reactions¹⁸ have already been reported.

Optically active amino alcohols **13**, which were derived from L-proline, were added in a catalytic amount (30 mol%) to activate the enolate equivalent **7** formed from dizinc **1** and phenyl isocyanate (**2b**) for the addition to *p*-tolylaldehyde (**5b**) (Scheme 4). In all cases, asymmetric induction was observed. Among the structures of **13**, (*S*)-bis(4-fluorophenyl)(1-methylpyrrolidin-2-yl)methanol (**13g**)¹⁹ induced the highest enantioselectivity (87% ee).

Table 4 Reformatsky-Type Reaction Starting from Dizinc **9** and Isocyanates **2**^a

Entry	R	T ₁ (°C)	T ₂ (°C)	Yield ^b (%) of 12	syn/anti
1	Bz (2a)	−60	−40	47 (12aa)	6:1
2	Ph (2b)	80	−40	<5 (12ba)	—
3	Ph (2b)	80	25	<5 (12ba)	—
4	Ph (2b)	80	140	<5 (12ba)	—

^a Reaction conditions: dizinc **9** (0.13 M in THF, 0.2 mmol), isocyanate **2** (0.2 mmol), solvent (4 mL), aldehyde **5a** (0.2 mmol).^b Isolated yields.

The reactions in Scheme 4 were performed under dilution conditions to ensure a slow reaction rate, as the milder conditions may benefit the asymmetric induction: in each case, the concentration of **5b** was 2.5 mM. More concentrated conditions were examined, which resulted in slightly lower enantioselectivities (Table 5, entries 1 and 2). As indicated in Table 2, successful asymmetric induction was also expected in the case of the less reactive enolate equivalent **7** formed from dizinc **1** and cyclohexyl isocyanate (**2e**); unfortunately, the yields and enantioselectivity obtained for the asymmetric Reformatsky-type reaction were inferior to those obtained for the enolate equivalent formed from **1** and phenyl isocyanate (**2b**) (Table 5, entries 5 and 6).

Optimization of the reaction parameters for the addition of the enolate equivalent **7** formed from dizinc **1** and phenyl isocyanate (**2b**) to *p*-tolylaldehyde (**5b**) in the presence of **13g** was examined (Table 6). When the reaction was performed by adding 2.5 equivalents of the enolate to **5b** in

the presence of 30 mol% **13g** for 72 hours, the adduct **6bb** was obtained in 99% yield with 84% ee (Table 6, entry 4). When less catalyst was used, the yield of the product was decreased (Table 6, entries 5 and 6).

As shown in Table 7, the reactions of this enolate equivalent with various aldehydes were examined using the conditions of entry 4 in Table 6. In all cases, the products **6bb–bn** were obtained with over 76% ee. Compared to previously reported catalytic asymmetric Reformatsky-type reactions, the present results show competitive optical yields.

The moderate reactivity of the enolate equivalent **7** can realize an asymmetric Reformatsky-type reaction with aldehydes even in the presence of a keto group. As shown in Scheme 5, treatment of keto aldehyde **14** with the enolate equivalent **7** formed from dizinc **1** and phenyl isocyanate (**2b**) in the presence of amino alcohol **13g** (30 mol%) afforded the corresponding product **15** in 81% yield with 93% ee; the keto group in the substrate remained intact.

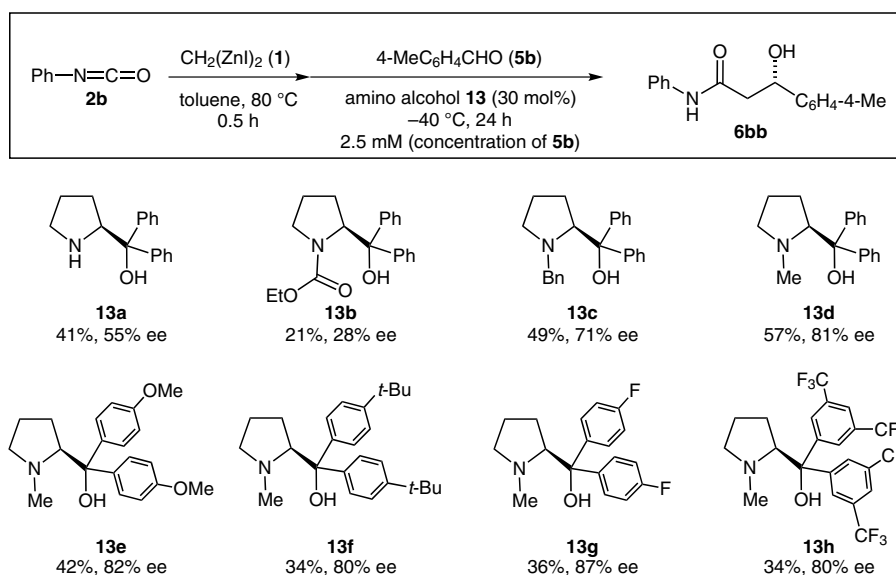
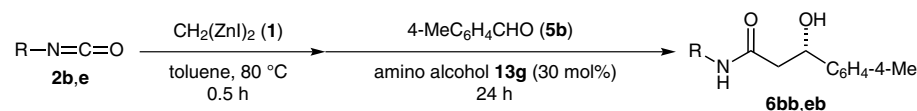
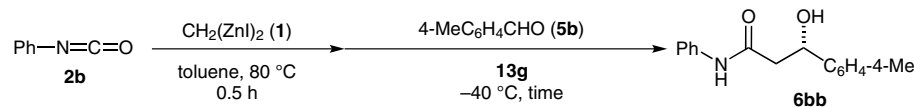
**Scheme 4** Asymmetric induction in the reaction of the enolate equivalent formed from **1** and **2b** with *p*-tolylaldehyde (**5b**) in the presence of L-proline-derived amino alcohols **13**

Table 5 Optimization of the Catalytic Asymmetric Reformatsky-Type Reaction Using the Enolate Equivalent Formed from **1** and **2b** or **2e** in the Presence of **13g**^a

Entry	R	Temp (°C)	Concn of 5b (mM)	Yield (%) of 6	ee (%)
1	Ph (2b)	−40	10.0	88 (6bb)	73
2	Ph (2b)	−40	5.0	59 (6bb)	79
3	Ph (2b)	−40	2.5	36 (6bb)	87
4	Ph (2b)	0	2.5	36 (6bb)	61
5	Cy (2e)	−40	2.5	5 (6eb)	45
6	Cy (2e)	0	2.5	3 (6eb)	60

^a Reaction conditions: **1** (0.13 M in THF, 1.5 mL, 0.2 mmol) and **2** (0.2 mmol) in toluene (2.0 mL) were premixed for 0.5 h at 80 °C; toluene (2.5 mL for entry 1, 12.5 mL for entry 2, 32.5 mL for entries 3–6), **13g** (0.03 mmol) in toluene (2.0 mL), and **5b** (0.1 mmol) in toluene (2.0 mL), were subsequently added at −40 °C.

Table 6 Optimization of the Reaction of the Enolate Equivalent Formed from **1** and **2b** with *p*-Tolylaldehyde (**5b**) in the Presence of a Catalytic Amount of **13g**^a

Entry	Equiv of 2b and 1	mol% of 13g	Time (h)	Yield (%) of 6bb	ee (%)
1	2.0	30	24	36	87
2	2.0	30	48	35	83
3	2.0	30	72	52	84
4	2.5	30	72	99	84
5	2.5	20	72	68	81
6	2.5	10	72	56	75

^a Reaction conditions: **1** (0.13 M in THF) and **2b** in toluene were premixed for 0.5 h at 80 °C; **13g** (0.03 mmol) in toluene, and **5b** (0.1 mmol) in toluene, were subsequently added at −40 °C. The concentration of **5b** was 2.5 mM.

In conclusion, the enolate equivalents prepared from isocyanates and dizinc compounds have only moderate nucleophilicity, which can be enhanced by the addition of a catalytic amount of a 2-amino alcohol. When the added 2-amino alcohol is optically active, moderate to good asymmetric induction is observed. In addition, the asymmetric

Reformatsky-type reaction with aldehydes can be performed chemoselectively, even in the presence of a keto group. This reaction of a dizinc compound with an isocyanate is a novel strategy for obtaining an amide enolate equivalent, which has unique reactivity. The reasonable catalytic asymmetric induction in these reactions was ac-

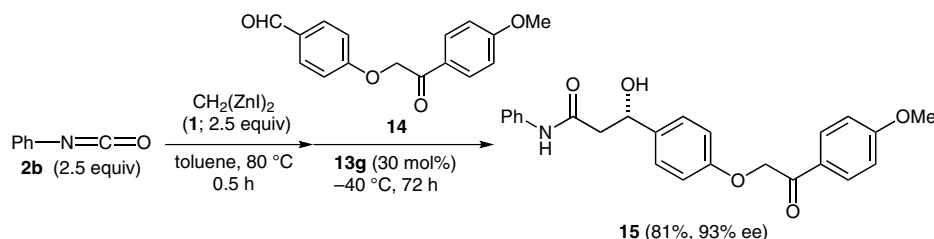
**Scheme 5** Asymmetric induction in the chemoselective Reformatsky-type reaction

Table 7 Examples of the Reaction of **7** Formed from **1** and **2b** with Aldehydes **5** in the Presence of a Catalytic Amount of **13g**^a

$\text{Ph-N=C=O} \xrightarrow[\text{toluene, 80 } ^\circ\text{C, 0.5 h}]{\text{CH}_2(\text{Zn})_2 \text{ (1, 2.5 equiv)}} \xrightarrow[\text{13g (30 mol\%), -40 } ^\circ\text{C, 72 h}]{\text{RCHO 5}} \text{Ph-NH-C(=O)-CH(OH)-R}$			
2b (2.5 equiv)			
Entry	R	Yield (%) of 6	ee ^b (%)
1	4-Tol	99 (6bb)	84
2	2-Tol	91 (6bc)	84
3	4-FC ₆ H ₄	77 (6bd)	89
4	4-ClC ₆ H ₄	76 (6be)	79
5	4-BrC ₆ H ₄	47 (6bf)	77
6	4-MeOC ₆ H ₄	65 (6bg)	94
7	4- <i>t</i> -BuC ₆ H ₄	57 (6bh)	84
8	Mes	63 (6bi)	84
9	2-Naph	89 (6bj)	88
10	2-thienyl	73 (6bk)	94
11	2-furyl	92 (6bl)	76
12	Me	45 (6bm)	76
13	<i>t</i> -Bu	49 (6bn)	83

^a Reaction conditions: **1** (0.13 M in THF, 2.0 mL, 0.25 mmol) and **2b** (0.25 mmol) in toluene (4.0 mL) were premixed for 0.5 h at 80 °C; toluene (32 mL), **13g** (0.03 mmol) in toluene (2.0 mL) and aldehyde **5** (0.1 mmol) in toluene (2.0 mL) were subsequently added at –40 °C.

^b The absolute configuration of the product in entry 12 was determined by comparison with an authentic sample.²⁰ The absolute configurations of the other products were deduced based on this result.

complished with an optically active Lewis base, not with an optically active Lewis acid.

Nuclear magnetic resonance spectra were recorded on a Varian Unity Inova 500 (¹H, 500 MHz; ¹³C, 125.7 MHz) spectrometer using TMS as an internal standard for ¹H NMR (δ = 0 ppm) and CDCl₃ as an internal standard for ¹³C NMR (δ = 77.0 ppm) measurements. ¹H NMR data are reported as follows: chemical shift, multiplicity (s = singlet, d = doublet, t = triplet, q = quartet, br = broad, m = multiplet), coupling constants (Hz), and integration. High-resolution mass spectra were obtained with a Thermo Fisher Scientific Exactive (ESI, APCI) mass spectrometer. Infrared (IR) spectra were determined on a Shimadzu FTIR-8200PC spectrometer. Melting points were determined using a Yanako MP-500D apparatus. TLC analyses were performed by means of Merck Kieselgel 60 F254 (0.25 mm) plates. Visualization was accomplished with UV light (254 nm) and an aqueous vanillin solution followed by heating. Flash column chromatography was carried out using Kanto Chemical Co. silica gel (spherical, 40–100 μm). Unless otherwise noted, commercially available reagents were used without purification. Tetrahydrofuran (dehydrated, stabilizer free, 'Super') was purchased from Kanto Chemical Co., stored under argon, and used as is. Zinc powder was used after washing with 10% HCl according to the reported procedure.²¹ The preparation methods and characteri-

zation of the chiral ligands **13**, and spectra (NMR, HPLC) of all compounds, are given in the Supporting Information.

Bis(iodozincio)methane (**1**)^{6a}

A mixture of pure zinc dust (150 mmol), diiodomethane (1.0 mmol), and PbCl₂ (0.005 mmol) in THF (5.0 mL) was sonicated for 1 h in an ultrasonic cleaner bath under argon. When pyrometallurgy zinc dust was used instead of pure zinc, it was not necessary to add PbCl₂. Both pure zinc and pyrometallurgy zinc are commercially available. Diiodomethane (50 mmol) in THF (45 mL) was added dropwise to the mixture over 30 min at 0 °C with vigorous stirring. The mixture was then stirred for 4 h at 25 °C. After the stirring was stopped, the reaction vessel was allowed to stand undisturbed for several hours. Excess zinc was separated by sedimentation. The ¹H NMR spectrum of the obtained supernatant showed a broad singlet at –1.2 ppm at 0 °C, which corresponds to the methylene protons of **1**. The supernatant was used in further reactions as a solution of **1** in THF (0.1–0.5 M). The concentration of **1** was estimated by ¹H NMR analysis using 2,2,3,3-tetramethylbutane as an internal standard. Bis(iodozincio)methane in THF can be kept for at least a month in a sealed reaction vessel.

1,1-Bis(iodozincio)ethane (**9**)¹³

A mixture of pure zinc dust (150 mmol), 1,1-diiodoethane (1.0 mmol), and PbCl₂ (0.005 mmol) in THF (5.0 mL) was sonicated for 1 h in an ultrasonic cleaner bath under argon. When pyrometallurgy zinc dust was used instead of pure zinc, it is not necessary to add PbCl₂. Both pure zinc and pyrometallurgy zinc are commercially available. To the mixture, 1,1-diiodoethane (50 mmol) in THF (45 mL) was added dropwise over 30 min at 0 °C with vigorous stirring. Then, the mixture was stirred for 4 h at 25 °C. After the stirring was stopped, the reaction vessel was allowed to stand undisturbed for several hours. Excess zinc was separated by sedimentation. The ¹H NMR spectrum of the obtained supernatant showed a quartet at –0.08 ppm at 20 °C, which corresponds to the methyne proton of **9**; ¹H NMR (300 MHz, 20 °C): δ = 1.45 (d, *J* = 7.8 Hz, 3 H), –0.08 (q, *J* = 7.8 Hz, 1 H). The supernatant was used in further reactions as a solution of **9** in THF (0.1–0.5 M). The concentration of **9** was estimated by ¹H NMR analysis using 2,2,3,3-tetramethylbutane as an internal standard. 1,1-Bis(iodozincio)ethane in THF can be kept unchanged for at least 2 days in a sealed reaction vessel.

N,N'-Dibenzoylmalonamide (**4a**)

To a solution of benzoyl isocyanate (**2a**; 29.4 mg, 0.2 mmol) in THF (4.0 mL), dizinc **1** (0.13 M in THF, 0.2 mmol) was added dropwise at 25 °C under argon. The reaction mixture was stirred for 30 min at the same temperature. The resulting mixture was poured into aq NH₄Cl (1 M, 10 mL). The mixture was extracted with EtOAc (2 × 30 mL). The combined organic layers were washed with brine (30 mL) and dried over Na₂SO₄. Purification by silica gel column chromatography (hexane–EtOAc, 1:2) gave **4a** [yield: 27.2 mg (88%); *N*-acetylbenzamide (**3a**) was also isolated in 6% yield (2.6 mg)].

Yield: 27.2 mg (88%); white solid; mp 181.0–182.0 °C.

TLC: *R*_f = 0.36 (hexane–EtOAc, 1:1).

IR (KBr): 3326.4, 3278.2, 2958.9, 2930.0, 1722.5, 1710.9, 1685.9, 1673.3, 1599.1, 1469.8, 1357.9, 1317.4, 1257.6, 1202.7, 1159.3, 706.9 cm^{–1}.

¹H NMR (CDCl₃): δ = 8.79 (s, 2 H), 7.87 (d, *J* = 7.8 Hz, 4 H), 7.63 (t, *J* = 7.8 Hz, 2 H), 7.52 (t, *J* = 7.8 Hz, 4 H), 4.62 (s, 2 H).

¹³C NMR (CDCl₃): δ = 168.6, 165.4, 133.5, 132.2, 129.1, 127.7, 48.1.

HRMS (ESI): *m/z* [M + H]⁺ calcd for C₁₇H₁₅N₂O₄: 311.1026; found: 311.1012.

N-(3-Hydroxy-3-phenylpropanoyl)benzamide (**6aa**)

To a solution of benzoyl isocyanate (**2a**; 29.4 mg, 0.2 mmol) in THF (4.0 mL), dizinc **1** (0.13 M in THF, 0.2 mmol) was added dropwise

at $-60\text{ }^{\circ}\text{C}$ under argon. The reaction mixture was stirred for 30 min at $-60\text{ }^{\circ}\text{C}$. After the reaction mixture was warmed to $-40\text{ }^{\circ}\text{C}$, benzaldehyde (**5a**; 20 μL , 0.2 mmol) was added to the resulting mixture. The mixture was stirred at $-40\text{ }^{\circ}\text{C}$ for 3 h, then poured into sat. aq NH_4Cl (10 mL). The mixture was extracted with EtOAc ($2 \times 30\text{ mL}$). The combined organic layers were washed with brine (30 mL) and dried over Na_2SO_4 . Purification by silica gel column chromatography (hexane–EtOAc, 1:1) gave compound **6aa**.

Yield: 48.4 mg (90%); white solid; mp $125.5\text{--}126.0\text{ }^{\circ}\text{C}$.

TLC: $R_f = 0.30$ (hexane–EtOAc, 1:1).

IR (KBr): 3475.9, 3286.8, 1700.3, 1668.5, 1510.3, 1469.8, 1375.3, 1304.9, 1248.0, 1183.4, 1032.0, 910.4, 770.6, 705.0 cm^{-1} .

^1H NMR (CDCl_3): $\delta = 9.03$ (s, 1 H), 7.85 (d, $J = 7.8\text{ Hz}$, 2 H), 7.62 (t, $J = 7.5\text{ Hz}$, 1 H), 7.50 (t, $J = 7.8\text{ Hz}$, 2 H), 7.43 (d, $J = 7.8\text{ Hz}$, 2 H), 7.37 (t, $J = 7.5\text{ Hz}$, 2 H), 7.31 (d, $J = 6.9\text{ Hz}$, 1 H), 5.27 (m, 1 H), 3.40 (m, 2 H).

^{13}C NMR (CDCl_3): $\delta = 174.6$, 165.5, 142.4, 133.5, 132.4, 129.0, 128.6, 127.8, 127.7, 125.8, 70.1, 46.5.

HRMS (ESI): m/z $[\text{M} + \text{Na}]^+$ calcd for $\text{C}_{16}\text{H}_{15}\text{NO}_3\text{Na}$: 292.0944; found: 292.0937.

Reformatsky-Type Reaction for the Formation of **6ba–6ea**; General Procedure

To a solution of isocyanate **2** (0.2 mmol) in THF or toluene (4.0 mL), dizinc **1** (0.13 M in THF, 0.2 mmol) was added dropwise at $80\text{ }^{\circ}\text{C}$ under argon. The reaction mixture was stirred for 30 min at $80\text{ }^{\circ}\text{C}$. Then, benzaldehyde (**5a**; 20 μL , 0.2 mmol) was added to the resulting mixture at $25\text{ }^{\circ}\text{C}$. The mixture was stirred at $25\text{ }^{\circ}\text{C}$ for 3 h, then poured into sat. aq NH_4Cl (10 mL). The mixture was extracted with EtOAc ($2 \times 30\text{ mL}$) and the combined organic layers were washed with brine (30 mL) and dried over Na_2SO_4 . Purification by silica gel column chromatography (hexane–EtOAc) gave compound **6ba–6ea**.

3-Hydroxy-*N*,3-diphenylpropanamide (**6ba**)

CAS Reg. No. [4198-15-6].

Yield: 9.6 mg (20%); white solid.

^1H NMR (CDCl_3): $\delta = 9.18$ (s, 1 H), 7.40 (dd, $J = 8.5$, 1.0 Hz, 2 H), 7.25 (d, $J = 8.5\text{ Hz}$, 2 H), 7.17 (t, $J = 7.0\text{ Hz}$, 2 H), 7.11 (m, 3 H), 6.90 (t, $J = 7.5\text{ Hz}$, 1 H), 5.08 (d, $J = 3.5\text{ Hz}$, 1 H), 4.99 (m, 1 H), 2.62 (dd, $J = 15.0$, 9.5 Hz, 1 H), 2.53 (dd, $J = 15.0$, 3.0 Hz, 1 H).

^{13}C NMR (CDCl_3): $\delta = 169.9$, 143.6, 138.0, 128.3, 128.0, 127.0, 125.3, 123.4, 119.6, 70.2, 45.7.

3-Hydroxy-*N*-(4-methoxyphenyl)-3-phenylpropanamide (**6ca**)

Yield: 11.9 mg (22%); white solid; mp $140.5\text{--}141.0\text{ }^{\circ}\text{C}$.

TLC: $R_f = 0.32$ (hexane–EtOAc, 1:1).

IR (KBr): 3287.8, 3253.1, 1656.0, 1604.8, 1552.8, 1511.3, 1414.9, 1303.0, 1248.0, 1181.5, 1065.7, 1035.8, 967.3, 832.3, 776.4, 757.1, 698.3 cm^{-1} .

^1H NMR (CDCl_3): $\delta = 8.92$ (s, 1 H), 7.34 (t, $J = 9.3\text{ Hz}$, 4 H), 7.25 (t, $J = 9.3\text{ Hz}$, 2 H), 7.17 (t, $J = 7.5\text{ Hz}$, 1 H), 6.74 (td, $J = 8.7$, 3.3 Hz, 2 H), 5.04 (m, 2 H), 3.69 (s, 3 H), 2.64 (m, 2 H).

^{13}C NMR (CDCl_3): $\delta = 169.9$, 155.9, 143.6, 131.1, 128.2, 127.2, 125.4, 121.6, 113.7, 70.5, 55.2, 45.5.

HRMS (ESI): m/z $[\text{M} + \text{Na}]^+$ calcd for $\text{C}_{16}\text{H}_{17}\text{NO}_3\text{Na}$: 294.1101; found: 294.1091.

3-Hydroxy-3-phenyl-*N*-[4-(trifluoromethyl)phenyl]propanamide (**6da**)

Yield: 24.1 mg (39%); white solid; mp $191.2\text{--}192.0\text{ }^{\circ}\text{C}$.

TLC: $R_f = 0.50$ (hexane–EtOAc, 1:1).

IR (KBr): 3335.1, 1661.8, 1603.9, 1532.5, 1456.3, 1411.0, 1331.9, 1252.8, 1151.6, 1113.0, 1071.5, 1019.4, 973.1, 912.4, 888.3, 866.1, 836.2, 758.1, 702.1 cm^{-1} .

^1H NMR (CDCl_3): $\delta = 9.56$ (s, 1 H), 7.44 (d, $J = 8.4\text{ Hz}$, 2 H), 7.20 (d, $J = 8.1\text{ Hz}$, 2 H), 7.10 (d, $J = 7.2\text{ Hz}$, 2 H), 7.02 (t, $J = 7.5\text{ Hz}$, 2 H), 6.93 (t, $J = 8.1\text{ Hz}$, 1 H), 5.00 (d, $J = 3.9\text{ Hz}$, 1 H), 4.85 (m, 1 H), 2.50 (dd, $J = 15.0$, 9.3 Hz, 1 H), 2.37 (dd, $J = 14.7$, 3.6 Hz, 1 H).

^{13}C NMR (CDCl_3): $\delta = 170.2$, 143.5, 141.4, 127.9, 127.0, 125.4 (q, $J = 14.4\text{ Hz}$), 125.2, 124.4, 123.4 (q, $J = 211.2\text{ Hz}$), 119.0, 70.0, 46.0.

^{19}F NMR (CDCl_3): $\delta = -62.3$.

HRMS (ESI): m/z $[\text{M} + \text{H}]^+$ calcd for $\text{C}_{16}\text{H}_{15}\text{F}_3\text{NO}_2$: 310.1049; found: 310.1036.

N-Cyclohexyl-3-hydroxy-3-phenylpropanamide (**6ea**)

Yield: 3.0 mg (6%); white solid; mp $129.2\text{--}129.5\text{ }^{\circ}\text{C}$.

TLC: $R_f = 0.43$ (hexane–EtOAc, 1:2).

IR (KBr): 3302.3, 3087.2, 3023.6, 2933.9, 2854.8, 1638.6, 1553.7, 1450.5, 1360.8, 1207.5, 1054.1, 1020.4, 984.7, 893.1, 698.3 cm^{-1} .

^1H NMR (CDCl_3): $\delta = 7.31$ (m, 5 H), 5.55 (d, $J = 7.2\text{ Hz}$, 1 H), 5.09 (m, 1 H), 4.31 (d, $J = 3.0\text{ Hz}$, 1 H), 3.77 (m, 1 H), 2.53 (m, 2 H), 1.88 (m, 2 H), 1.65 (m, 3 H), 1.33 (m, 2 H), 1.10 (m, 3 H).

^{13}C NMR (CDCl_3): $\delta = 170.8$, 143.0, 128.5, 127.6, 125.6, 70.9, 48.2, 44.7, 33.0, 25.4, 24.8.

HRMS (ESI): m/z $[\text{M} + \text{Na}]^+$ calcd for $\text{C}_{15}\text{H}_{21}\text{NO}_2\text{Na}$: 270.1465; found: 270.1456.

N-(3-Hydroxy-2-methyl-3-phenylpropanoyl)benzamide (**12aa**)

To a solution of benzoyl isocyanate (**2a**; 29.4 mg, 0.2 mmol) in THF (4.0 mL), dizinc **9** (0.13 M in THF, 0.2 mmol) was added dropwise at $-60\text{ }^{\circ}\text{C}$ under argon. The reaction mixture was stirred for 30 min at $-60\text{ }^{\circ}\text{C}$. After the reaction mixture was warmed to $-40\text{ }^{\circ}\text{C}$, benzaldehyde (**5a**; 20 μL , 0.2 mmol) was added to the resulting mixture. The mixture was stirred at $-40\text{ }^{\circ}\text{C}$ for 3 h, then poured into sat. aq NH_4Cl (10 mL). The mixture was extracted with EtOAc ($2 \times 30\text{ mL}$). The combined organic layers were washed with brine (30 mL) and dried over Na_2SO_4 . Purification by silica gel column chromatography (hexane–EtOAc, 3:2) gave compound **12aa** as a diastereomeric mixture (*syn/anti* = 6:1).¹⁴

Yield: 26.6 mg (47%); colorless oil.

TLC: $R_f = 0.50$ (hexane–EtOAc, 1:2).

IR (neat): 3410.3, 2981.1, 2937.7, 1730.2, 1680.1, 1601.0, 1507.4, 1484.3, 1369.5, 1264.4, 1159.3, 1121.7, 1027.1, 701.2, 665.5 cm^{-1} .

^1H NMR (CDCl_3): $\delta = 9.07$ (br s, 1 H), 7.85 (m, 2 H), 7.61 (t, $J = 7.2\text{ Hz}$, 1 H), 7.50 (t, $J = 7.5\text{ Hz}$, 2 H), 7.34 (m, 5 H), 5.24 (dd, $J = 4.0$, 3.0 Hz, 0.86 H), 4.85 (dd, $J = 8.0$, 6.0 Hz, 0.14 H), 3.75 (m, 1 H), 3.15 (d, $J = 8.0\text{ Hz}$, 0.14 H), 3.10 (d, $J = 3.0\text{ Hz}$, 0.86 H), 1.19 (d, $J = 6.9\text{ Hz}$, 3 H).

^{13}C NMR (CDCl_3): δ (major isomer) = 177.8, 165.2, 141.0, 133.3, 132.8, 129.0, 128.3, 127.7, 127.6, 126.0, 73.3, 47.2, 10.2.

HRMS (ESI): m/z $[\text{M} + \text{H}]^+$ calcd for $\text{C}_{17}\text{H}_{18}\text{NO}_3$: 284.1281; found: 284.1268.

Asymmetric Synthesis of β -Hydroxy Amides **6bb–6bn** and **15**; General Procedure

To a solution of phenyl isocyanate (**2b**; 30.0 mg, 0.25 mmol) in toluene (4.0 mL), dizinc **1** (0.13 M in THF, 2.0 mL, 0.25 mmol) was added dropwise at $80\text{ }^{\circ}\text{C}$ under argon. After being stirred for 30 min at $80\text{ }^{\circ}\text{C}$, the mixture was diluted with toluene (32 mL) at $-40\text{ }^{\circ}\text{C}$. To the resulting mixture, (*S*)-bis(4-fluorophenyl)(1-methylpyrrolidin-2-yl)methanol (**13g**; 9.1 mg, 0.03 mmol) in toluene (2.0 mL) and an aldehyde **5** (0.1 mmol) in toluene (2.0 mL) were added. The resulting mixture was stirred at $-40\text{ }^{\circ}\text{C}$ for 72 h, then poured into aq HCl (1 M, 10 mL). The mixture was extracted with EtOAc (2×30

mL) and the combined organic layers were washed with brine (30 mL) and dried over Na_2SO_4 . Purification by silica gel column chromatography (hexane–EtOAc, 1:1) gave the adducts **6bb–6bn** efficiently.²⁰ The chemoselective Reformatsky-type reaction from **14** to **15** was also performed according to this procedure.

(S)-3-Hydroxy-N-phenyl-3-(p-tolyl)propanamide (6bb)

Yield: 25.2 mg (99%); 84% ee; white solid; mp 181.0–182.0 °C.

$[\alpha]_{\text{D}}^{20}$ –83.3 (*c* 0.06, CH_2Cl_2).

TLC: R_f = 0.45 (hexane–EtOAc, 1:1).

HPLC (Daicel Chiralcel AD-H, hexane–*i*-PrOH, 20:1, flow rate = 2.0 mL/min, λ = 254 nm, 40 °C): t_{R} (minor) = 17.6 min, t_{R} (major) = 20.0 min.

IR (KBr): 3330.2, 1657.9, 1601.0, 1540.2, 1498.8, 1442.8, 1363.7, 1064.8, 815.9, 752.3, 691.5, 502.5 cm^{-1} .

^1H NMR (CDCl_3): δ = 8.86 (s, 1 H), 7.50 (d, J = 9.0 Hz, 2 H), 7.26 (m, 4 H), 7.12 (d, J = 7.5 Hz, 2 H), 7.04 (t, J = 7.5 Hz, 1 H), 5.09 (m, 1 H), 4.68 (d, J = 3.0 Hz, 1 H), 2.74 (dd, J = 15.0, 9.5 Hz, 1 H), 2.65 (dd, J = 15.5, 3.0 Hz, 1 H), 2.29 (s, 3 H).

^{13}C NMR (CDCl_3): δ = 170.1, 140.5, 138.1, 137.1, 129.0, 128.7, 125.5, 123.9, 119.9, 70.5, 45.9, 21.0.

HRMS (ESI): m/z $[\text{M} + \text{Cl}]^-$ calcd for $\text{C}_{16}\text{H}_{17}\text{NO}_2\text{Cl}$: 290.0942; found: 290.0956.

(S)-3-Hydroxy-N-phenyl-3-(o-tolyl)propanamide (6bc)

Yield: 23.2 mg (91%); 84% ee; white solid; mp 161.2–162.0 °C.

$[\alpha]_{\text{D}}^{20}$ –41.7 (*c* 0.12, CH_2Cl_2).

TLC: R_f = 0.53 (hexane–EtOAc, 1:1).

HPLC (Daicel Chiralpak IA, hexane–*i*-PrOH, 20:1, flow rate = 2.0 mL/min, λ = 254 nm, 40 °C): t_{R} (minor) = 14.0 min, t_{R} (major) = 16.8 min.

IR (KBr): 3304.2, 1659.8, 1599.1, 1540.2, 1499.7, 1444.8, 1360.8, 1312.6, 1300.1, 1066.7, 1022.3, 756.1, 729.1, 693.4, 498.6 cm^{-1} .

^1H NMR (CDCl_3): δ = 7.76 (s, 1 H), 7.54 (m, 3 H), 7.34 (t, J = 8.0 Hz, 2 H), 7.26 (m, 1 H), 7.21 (dt, J = 1.5, 8.5 Hz, 1 H), 7.14 (m, 2 H), 5.45 (td, J = 2.5, 10.0 Hz, 1 H), 3.31 (d, J = 2.5 Hz, 1 H), 2.76 (dd, J = 15.0, 10.0 Hz, 1 H), 2.65 (dd, J = 16.0, 2.5 Hz, 1 H), 2.38 (s, 3 H).

^{13}C NMR (CDCl_3): δ = 169.8, 140.8, 137.6, 134.2, 130.6, 129.1, 127.8, 126.6, 125.1, 124.5, 120.1, 67.7, 44.9, 19.0.

HRMS (ESI): m/z $[\text{M} + \text{Cl}]^-$ calcd for $\text{C}_{16}\text{H}_{17}\text{NO}_2\text{Cl}$: 290.0942; found: 290.0954.

(S)-3-(4-Fluorophenyl)-3-hydroxy-N-phenylpropanamide (6bd)

Yield: 20.0 mg (77%); 89% ee; white solid; mp 169.0–170.0 °C.

$[\alpha]_{\text{D}}^{20}$ –20.8 (*c* 0.24, CH_2Cl_2).

TLC: R_f = 0.32 (hexane–EtOAc, 1:1).

HPLC (Daicel Chiralcel AD-H, hexane–*i*-PrOH, 20:1, flow rate = 2.0 mL/min, λ = 254 nm, 40 °C): t_{R} (major) = 16.0 min, t_{R} (minor) = 19.0 min.

IR (KBr): 3335.1, 1654.0, 1602.9, 1540.2, 1512.3, 1499.7, 1444.8, 1227.7, 831.4, 756.1, 691.5 cm^{-1} .

^1H NMR (CDCl_3): δ = 8.43 (s, 1 H), 7.51 (d, J = 8.5 Hz, 2 H), 7.38 (dt, J = 7.0, 1.5 Hz, 2 H), 7.30 (dt, J = 7.5, 2.0 Hz, 2 H), 7.09 (t, J = 7.5 Hz, 1 H), 7.03 (dt, J = 8.5, 1.5 Hz, 2 H), 5.16 (d, J = 8.5 Hz, 1 H), 4.50 (s, 1 H), 2.72 (m, 2 H).

^{13}C NMR (CDCl_3): δ = 170.0, 139.1, 137.9, 128.9, 127.3 (d, J = 8.3 Hz), 124.2, 120.0, 119.9, 115.3 (d, J = 21.4 Hz), 70.1, 45.9.

HRMS (ESI): m/z $[\text{M} + \text{Cl}]^-$ calcd for $\text{C}_{15}\text{H}_{14}\text{FNO}_2\text{Cl}$: 294.0692; found: 294.0706.

(S)-3-(4-Chlorophenyl)-3-hydroxy-N-phenylpropanamide (6be)

Yield: 21.0 mg (76%); 79% ee; white solid; mp 188.5–189.2 °C.

$[\alpha]_{\text{D}}^{20}$ –8.3 (*c* 0.03, CH_2Cl_2).

TLC: R_f = 0.46 (hexane–EtOAc, 1:1).

HPLC (Daicel Chiralcel AD-H, hexane–*i*-PrOH, 20:1, flow rate = 2.0 mL/min, λ = 254 nm, 40 °C): t_{R} (minor) = 17.2 min, t_{R} (major) = 22.8 min.

IR (KBr): 3328.3, 1656.0, 1602.0, 1540.2, 1499.7, 1444.8, 1425.5, 1369.5, 1092.7, 1067.7, 1012.7, 824.6, 756.1, 744.6, 692.5, 500.6 cm^{-1} .

^1H NMR (CDCl_3): δ = 7.50 (s, 1 H), 7.49 (d, J = 8.5 Hz, 2 H), 7.34 (m, 6 H), 7.14 (t, J = 7.0 Hz, 1 H), 5.21 (m, 1 H), 2.76 (dd, J = 15.0, 9.0 Hz, 1 H), 2.70 (dd, J = 15.0, 3.0 Hz, 1 H).

^{13}C NMR (CDCl_3): δ = 169.9, 137.8, 128.9, 128.6, 127.0, 124.3, 120.0, 119.9, 109.7, 70.1, 45.7.

HRMS (ESI): m/z $[\text{M} - \text{H}]^-$ calcd for $\text{C}_{15}\text{H}_{13}\text{ClNO}_2$: 274.0640; found: 274.0647.

(S)-3-(4-Bromophenyl)-3-hydroxy-N-phenylpropanamide (6bf)

Yield: 15.0 mg (47%); 77% ee; white solid; mp 186.0–186.6 °C.

$[\alpha]_{\text{D}}^{20}$ +25.0 (*c* 0.10, CH_2Cl_2).

TLC: R_f = 0.44 (hexane–EtOAc, 1:1).

HPLC (Daicel Chiralcel AD-H, hexane–*i*-PrOH, 20:1, flow rate = 2.0 mL/min, λ = 254 nm, 40 °C): t_{R} (minor) = 41.1 min, t_{R} (major) = 48.4 min.

IR (KBr): 3327.4, 1654.0, 1601.0, 1540.2, 1499.7, 1490.1, 1444.8, 1067.7, 1009.8, 821.7, 692.5 cm^{-1} .

^1H NMR (CDCl_3): δ = 9.06 (s, 1 H), 7.46 (dd, J = 8.0, 1.0 Hz, 2 H), 7.38 (d, J = 8.0 Hz, 2 H), 7.22 (m, 4 H), 7.00 (tt, J = 7.0, 1.0 Hz, 1 H), 5.13 (d, J = 3.5 Hz, 1 H), 5.04 (m, 1 H), 2.63 (m, 2 H).

^{13}C NMR (CDCl_3): δ = 169.8, 142.7, 138.0, 131.3, 128.6, 127.3, 123.9, 121.0, 119.9, 69.9, 45.6.

HRMS (ESI): m/z $[\text{M} + \text{Cl}]^-$ calcd for $\text{C}_{15}\text{H}_{14}\text{BrNO}_2\text{Cl}$: 353.9891; found: 353.9911.

(S)-3-Hydroxy-3-(4-methoxyphenyl)-N-phenylpropanamide (6bg)

Yield: 17.6 mg (65%); 94% ee; white solid; mp 185.0–186.0 °C.

$[\alpha]_{\text{D}}^{20}$ –35.7 (*c* 0.07, CH_2Cl_2).

TLC: R_f = 0.30 (hexane–EtOAc, 1:1).

HPLC (Daicel Chiralcel AD-H, hexane–*i*-PrOH, 20:1, flow rate = 2.0 mL/min, λ = 254 nm, 40 °C): t_{R} (minor) = 37.3 min, t_{R} (major) = 41.9 min.

IR (KBr): 3325.4, 1658.9, 1601.0, 1538.3, 1515.1, 1499.7, 1443.8, 1365.7, 1300.1, 1178.6, 1065.7, 1037.8, 829.5, 753.2, 692.5, 502.5 cm^{-1} .

^1H NMR (CDCl_3): δ = 8.96 (s, 1 H), 7.49 (d, J = 7.5 Hz, 2 H), 7.28 (m, 4 H), 7.26 (d, J = 7.5 Hz, 1 H), 6.82 (d, J = 9.0 Hz, 2 H), 5.06 (td, J = 3.0, 9.5 Hz, 1 H), 4.79 (d, J = 2.5 Hz, 1 H), 3.74 (s, 3 H), 2.73 (dd, J = 15.5, 9.5 Hz, 1 H), 2.62 (dd, J = 15.0, 3.0 Hz, 1 H).

^{13}C NMR (CDCl_3): δ = 170.2, 158.9, 138.1, 135.7, 128.7, 126.8, 123.8, 119.9, 113.7, 70.2, 55.1, 45.9.

HRMS (ESI): m/z $[\text{M} - \text{H}]^-$ calcd for $\text{C}_{16}\text{H}_{16}\text{NO}_3$: 270.1136; found: 270.1142.

(S)-3-(4-tert-Butylphenyl)-3-hydroxy-N-phenylpropanamide (6bh)

Yield: 17.0 mg (57%); 84% ee; white solid; mp 110.0–111.0 °C.

$[\alpha]_{\text{D}}^{20}$ –25.0 (*c* 0.20, CH_2Cl_2).

TLC: R_f = 0.61 (hexane–EtOAc, 1:1).

HPLC (Daicel Chiralcel AD-H, hexane-*i*-PrOH, 20:1, flow rate = 2.0 mL/min, λ = 254 nm, 40 °C): t_R (major) = 19.3 min, t_R (minor) = 23.3 min.

IR (KBr): 3298.4, 2961.8, 1669.5, 1603.9, 1554.7, 1499.7, 1442.8, 1312.6, 1060.9, 819.8, 754.2, 690.6, 581.6 cm^{-1} .

^1H NMR (CDCl_3): δ = 7.80 (s, 1 H), 7.50 (d, J = 8.5 Hz, 2 H), 7.40 (d, J = 8.5 Hz, 2 H), 7.33 (m, 4 H), 7.13 (t, J = 7.5 Hz, 1 H), 5.20 (m, 1 H), 3.35 (d, J = 2.5 Hz, 1 H), 2.83 (dd, J = 15.5, 9.5 Hz, 1 H), 2.71 (dd, J = 15.0, 3.0 Hz, 1 H), 1.32 (s, 9 H).

^{13}C NMR (CDCl_3): δ = 169.8, 151.0, 139.7, 137.5, 129.0, 125.6, 125.3, 124.5, 120.0, 70.9, 46.0, 34.6, 31.3.

HRMS (ESI): m/z [$\text{M} + \text{Cl}$] $^-$ calcd for $\text{C}_{19}\text{H}_{23}\text{NO}_2\text{Cl}$: 332.1412; found: 332.1425.

(S)-3-Hydroxy-3-mesityl-*N*-phenylpropanamide (6bi)

Yield: 17.9 mg (63%); 84% ee; white solid; mp 110.0–111.0 °C.

$[\alpha]_D^{20}$ –16.9 (c 0.74, CH_2Cl_2).

TLC: R_f = 0.29 (hexane–EtOAc, 3:1).

HPLC (Daicel Chiralcel AD-H, hexane-*i*-PrOH, 20:1, flow rate = 2.0 mL/min, λ = 254 nm, 40 °C): t_R (major) = 8.5 min, t_R (minor) = 15.0 min.

IR (KBr): 3431.5, 3269.5, 2921.3, 1628.0, 1600.0, 1545.1, 1498.8, 1444.8, 1312.6, 1253.8, 1172.8, 1071.5, 850.6, 752.3, 690.6, 351.1 cm^{-1} .

^1H NMR (CDCl_3): δ = 8.05 (s, 1 H), 7.53 (d, J = 8.5 Hz, 2 H), 7.33 (t, J = 8.5 Hz, 2 H), 7.12 (t, J = 7.0 Hz, 1 H), 6.83 (s, 2 H), 5.65 (d, J = 10.5 Hz, 1 H), 3.14 (dd, J = 15.5, 5.5 Hz, 1 H), 3.10 (s, 1 H), 2.49 (dd, J = 15.5, 2.0 Hz, 1 H), 2.43 (s, 6 H), 2.26 (s, 3 H).

^{13}C NMR (CDCl_3): δ = 170.0, 137.7, 137.2, 136.0, 134.8, 130.3, 129.0, 124.4, 120.0, 68.2, 42.7, 20.7, 20.7.

HRMS (ESI): m/z [$\text{M} - \text{H}$] $^-$ calcd for $\text{C}_{18}\text{H}_{20}\text{NO}_2$: 282.1500; found: 282.1487.

(S)-3-Hydroxy-3-(2-naphthyl)-*N*-phenylpropanamide (6bj)

Yield: 25.9 mg (89%); 88% ee; white solid; mp 173.0–174.0 °C.

$[\alpha]_D^{20}$ –250.0 (c 0.02, CH_2Cl_2).

TLC: R_f = 0.44 (hexane–EtOAc, 1:1).

HPLC (Daicel Chiralcel AD-H, hexane-*i*-PrOH, 20:1, flow rate = 2.0 mL/min, λ = 254 nm, 40 °C): t_R (major) = 30.7 min, t_R (minor) = 39.2 min.

IR (KBr): 3299.4, 1664.6, 1602.0, 1549.9, 1499.7, 1488.2, 1445.7, 1363.7, 1314.5, 1068.6, 761.0, 744.6 cm^{-1} .

^1H NMR (CDCl_3): δ = 8.71 (s, 1 H), 7.87 (s, 1 H), 7.81 (m, 3 H), 7.52 (t, J = 8.5 Hz, 3 H), 7.45 (t, J = 7.5 Hz, 2 H), 7.29 (t, J = 7.0 Hz, 2 H), 7.07 (t, J = 7.5 Hz, 1 H), 5.33 (d, J = 9.5 Hz, 1 H), 4.79 (s, 1 H), 2.85 (dd, J = 16.0, 8.0 Hz, 1 H), 2.78 (dd, J = 15.5, 3.0 Hz, 1 H).

^{13}C NMR (CDCl_3): δ = 170.1, 140.8, 133.2, 132.9, 128.8, 128.3, 127.9, 127.6, 126.1, 125.8, 124.3, 124.1, 123.8, 121.1, 120.0, 70.9, 45.9.

HRMS (ESI): m/z [$\text{M} + \text{Cl}$] $^-$ calcd for $\text{C}_{19}\text{H}_{17}\text{NO}_2\text{Cl}$: 326.0942; found: 326.0954.

(S)-3-Hydroxy-*N*-phenyl-3-(2-thienyl)propanamide (6bk)

Yield: 18.1 mg (73%); 94% ee; white solid; mp 159.5–160.2 °C.

$[\alpha]_D^{20}$ –41.7 (c 0.12, CH_2Cl_2).

TLC: R_f = 0.50 (hexane–EtOAc, 1:1).

HPLC (Daicel Chiralcel AD-H, hexane-*i*-PrOH, 20:1, flow rate = 2.0 mL/min, λ = 254 nm, 40 °C): t_R (minor) = 18.9 min, t_R (major) = 21.4 min.

IR (KBr): 3297.5, 1679.1, 1663.7, 1607.7, 1599.1, 1554.7, 1498.8, 1444.8, 1319.4, 1254.8, 1236.4, 1086.0, 1046.4, 758.1, 700.2, 689.6 cm^{-1} .

^1H NMR (CDCl_3): δ = 7.64 (s, 1 H), 7.50 (d, J = 4.0 Hz, 2 H), 7.34 (t, J = 8.0 Hz, 2 H), 7.28 (d, J = 3.0 Hz, 1 H), 7.14 (t, J = 8.0 Hz, 1 H), 7.03 (d, J = 2.0 Hz, 1 H), 6.98 (t, J = 2.5 Hz, 1 H), 5.49 (m, 1 H), 3.79 (d, J = 3.5 Hz, 1 H), 2.90 (m, 2 H).

^{13}C NMR (CDCl_3): δ = 169.3, 146.3, 137.2, 129.1, 126.8, 125.0, 124.7, 123.8, 120.1, 67.2, 45.8.

HRMS (ESI): m/z [$\text{M} + \text{Cl}$] $^-$ calcd for $\text{C}_{13}\text{H}_{13}\text{NO}_2\text{SCl}$: 282.0350; found: 282.0367.

(S)-3-(2-Furyl)-3-hydroxy-*N*-phenylpropanamide (6bl)

Yield: 21.2 mg (92%); 76% ee; white solid; mp 133.0–134.0 °C.

$[\alpha]_D^{20}$ –31.3 (c 0.24, CH_2Cl_2).

TLC: R_f = 0.43 (hexane–EtOAc, 1:1).

HPLC (Daicel Chiralcel AD-H, hexane-*i*-PrOH, 20:1, flow rate = 2.0 mL/min, λ = 254 nm, 40 °C): t_R (minor) = 19.6 min, t_R (major) = 22.8 min.

IR (KBr): 3284.9, 1669.5, 1608.7, 1559.5, 1491.0, 1445.7, 1314.5, 1168.0, 1060.9, 1018.5, 759.0, 744.6, 693.4, 598.0 cm^{-1} .

^1H NMR (CDCl_3): δ = 7.71 (s, 1 H), 7.50 (d, J = 8.5 Hz, 2 H), 7.40 (m, 1 H), 7.34 (t, J = 7.0 Hz, 2 H), 7.13 (t, J = 7.5 Hz, 1 H), 6.35 (m, 1 H), 6.33 (d, J = 3.0 Hz, 1 H), 5.24 (m, 1 H), 3.62 (d, J = 4.5 Hz, 1 H), 2.96 (dd, J = 16.0, 8.5 Hz, 1 H), 2.87 (dd, J = 15.5, 3.5 Hz, 1 H).

^{13}C NMR (CDCl_3): δ = 169.4, 154.6, 142.3, 137.3, 129.0, 124.7, 120.1, 110.4, 106.5, 64.7, 42.0.

HRMS (ESI): m/z [$\text{M} - \text{H}$] $^-$ calcd for $\text{C}_{13}\text{H}_{12}\text{NO}_3$: 230.0823; found: 230.0812.

(R)-3-Hydroxy-*N*-phenylbutanamide (6bm)

Yield: 8.06 mg (45%); 76% ee; white solid; mp 102.8–103.5 °C.

$[\alpha]_D^{20}$ –21.2 (c 0.70, CHCl_3).

TLC: R_f = 0.28 (hexane–EtOAc, 1:2).

HPLC (Daicel Chiralcel AD-H, hexane-*i*-PrOH, 33:1, flow rate = 1.0 mL/min, λ = 254 nm, 40 °C): t_R (major) = 34.7 min, t_R (minor) = 38.8 min.

IR (KBr): 3250.2, 1662.7, 1601.0, 1553.7, 1500.7, 1445.7, 1338.7, 1319.4, 1128.4, 1064.8, 753.2, 694.4 cm^{-1} .

^1H NMR (CDCl_3): δ = 7.72 (s, 1 H), 7.50 (d, J = 9.0 Hz, 2 H), 7.33 (t, J = 8.0 Hz, 2 H), 7.12 (t, J = 7.5 Hz, 1 H), 4.32 (m, 1 H), 3.17 (d, J = 3.0 Hz, 1 H), 2.55 (dd, J = 3.0, 15.5 Hz, 1 H), 2.48 (dd, J = 9.0, 15.5 Hz, 1 H), 1.30 (d, J = 6.5 Hz, 3 H).

^{13}C NMR (CDCl_3): δ = 170.3, 137.5, 129.0, 124.5, 120.0, 65.0, 45.2, 23.0.

HRMS (ESI): m/z [$\text{M} + \text{Cl}$] $^-$ calcd for $\text{C}_{10}\text{H}_{13}\text{NO}_2\text{Cl}$: 214.0629; found: 214.0643.

(S)-3-Hydroxy-4,4-dimethyl-*N*-phenylpentanamide (6bn)

Yield: 8.8 mg (49%); 83% ee; yellow solid; mp 110.0–111.0 °C.

$[\alpha]_D^{20}$ –27.8 (c 0.27, CH_2Cl_2).

TLC: R_f = 0.61 (hexane–EtOAc, 1:1).

HPLC (Daicel Chiralcel AD-H, hexane-*i*-PrOH, 33:1, flow rate = 2.0 mL/min, λ = 254 nm, 40 °C): t_R (minor) = 13.0 min, t_R (major) = 18.3 min.

IR (KBr): 3293.6, 2958.9, 1659.3, 1603.9, 1552.8, 1501.7, 1445.7, 1337.7, 1149.6, 1070.5, 754.2, 702.1, 419.5 cm^{-1} .

^1H NMR (CDCl_3): δ = 7.86 (s, 1 H), 7.50 (d, J = 7.5 Hz, 2 H), 7.32 (t, J = 7.0 Hz, 2 H), 7.11 (t, J = 7.5 Hz, 1 H), 3.79 (d, J = 5.0 Hz, 1 H), 3.01 (s, 1 H), 2.54 (dd, J = 2.0, 15.5 Hz, 1 H), 2.43 (dd, J = 10.5, 15.5 Hz, 1 H), 0.96 (s, 9 H).

^{13}C NMR (CDCl_3): δ = 177.1, 137.7, 129.0, 124.4, 120.0, 76.4, 39.2, 34.7, 25.5.

HRMS (ESI): m/z $[\text{M} - \text{H}]^-$ calcd for $\text{C}_{13}\text{H}_{18}\text{NO}_2$: 220.1343; found: 220.1347.

(S)-3-Hydroxy-3-{4-[2-(4-methoxyphenyl)-2-oxoethoxy]phenyl}-N-phenylpropanamide (15)

Yield: 32.8 mg (81%); 93% ee; white solid; mp 142.5–143.0 °C.

$[\alpha]_{\text{D}}^{20} +10.0$ (c 0.25, CH_2Cl_2).

TLC: R_f = 0.38 (hexane–EtOAc, 1:2).

IR (KBr): 3514.5, 3265.6, 2920.3, 1681.0, 1602.0, 1541.2, 1445.7, 1320.3, 1256.7, 1219.1, 1171.8, 978.0, 832.3, 758.1 cm^{-1} .

^1H NMR (CDCl_3): δ = 8.00 (d, J = 9.0 Hz, 2 H), 7.73 (s, 1 H), 7.50 (d, J = 8.0 Hz, 2 H), 7.33 (m, 3 H), 7.12 (t, J = 7.5 Hz, 1 H), 7.13 (t, J = 7.5 Hz, 1 H), 6.97 (d, J = 8.5 Hz, 2 H), 6.93 (d, J = 8.5 Hz, 2 H), 5.23 (s, 2 H), 5.16 (d, J = 9.5 Hz, 1 H), 3.89 (s, 3 H), 3.45 (d, J = 2.5 Hz, 1 H), 2.78 (dd, J = 15.5, 9.5 Hz, 1 H), 2.67 (dd, J = 15.5, 3.0 Hz, 1 H).

^{13}C NMR (CDCl_3): δ = 193.0, 169.7, 164.1, 157.9, 137.5, 135.9, 130.6, 129.0, 127.6, 127.0, 124.5, 120.1, 115.0, 114.1, 70.8, 70.6, 55.5, 46.1.

HRMS (ESI): m/z $[\text{M} + \text{Cl}]^-$ calcd for $\text{C}_{24}\text{H}_{23}\text{NO}_5\text{Cl}$: 440.1259; found: 440.1276.

The enantiomeric excess of **15** was determined by HPLC analysis, after **15** was transformed into the corresponding *p*-bromobenzoyl ester **16**.

(S)-3-Anilino-1-{4-[2-(4-methoxyphenyl)-2-oxoethoxy]phenyl}-3-oxopropyl 4-Bromobenzoate (16)

To a solution of **15** (26.4 mg, 0.065 mmol) in CH_2Cl_2 (1.0 mL) was added 4-bromobenzoyl chloride (35.1 mg, 0.16 mmol), Et_3N (26.4 μL , 0.19 mmol), and DMAP (1.6 mg, 0.013 mmol) at 25 °C. After being stirred for 40 min at 25 °C, the resulting mixture was poured into sat. aq. NaHCO_3 (30 mL) and extracted with CH_2Cl_2 (2×30 mL). The combined organic layers were dried over Na_2SO_4 . Purification by silica gel column chromatography (hexane–EtOAc, 1:1) gave compound **16**.

Yield: 19 mg (50%); white solid; mp 129.0–130.0 °C.

$[\alpha]_{\text{D}}^{20} -250.7$ (c 0.04, CH_2Cl_2).

TLC: R_f = 0.43 (hexane–EtOAc, 1:1).

HPLC (Daicel Chiralcel OD-H, hexane–*i*-PrOH, 5.7:1, flow rate = 2.0 mL/min, λ = 254 nm, 40 °C): t_{R} (minor) = 68.2 min, t_{R} (major) = 79.7 min.

IR (KBr): 3342.8, 2965.7, 2932.9, 2361.9, 1717.7, 1684.9, 1656.9, 1602.0, 1513.2, 1441.9, 1267.3, 1217.1, 1171.8, 1117.8, 1104.3, 1007.9, 977.0, 829.4, 757.1, 695.4, 597.0, 545.9 cm^{-1} .

^1H NMR (CDCl_3): δ = 8.56 (m, 1 H), 7.93 (dd, J = 9.0, 1.5 Hz, 2 H), 7.84 (d, J = 8.5 Hz, 2 H), 7.49 (d, J = 8.5 Hz, 2 H), 7.44 (d, J = 7.5 Hz, 2 H), 7.36 (d, J = 9.0 Hz, 2 H), 7.22 (t, J = 7.5 Hz, 2 H), 7.01 (t, J = 7.5 Hz, 1 H), 6.92 (d, J = 9.0 Hz, 2 H), 6.87 (d, J = 9.0 Hz, 2 H), 6.39 (d, J = 7.0 Hz, 1 H), 5.17 (s, 2 H), 3.84 (s, 3 H), 3.10 (dd, J = 15.0, 9.0 Hz, 1 H), 2.86 (dd, J = 15.0, 5.5 Hz, 1 H).

^{13}C NMR (CDCl_3): δ = 192.7, 167.2, 164.6, 164.0, 158.0, 138.0, 132.5, 131.6, 131.1, 130.4, 129.0, 128.7, 128.0, 127.8, 127.4, 124.0, 119.9, 114.9, 114.0, 73.3, 70.5, 55.4, 44.3.

HRMS (ESI): m/z $[\text{M} + \text{Cl}]^-$ calcd for $\text{C}_{31}\text{H}_{26}\text{BrNO}_6\text{Cl}$: 622.0627; found: 622.0650.

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